

LECTURE 8: DIRECTED ACYCLICAL GRAPHS

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Economics 272, GSB 507, Spring 2025

OUTLINE

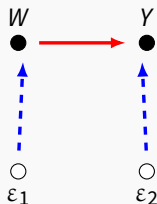
1. Directed Acyclical Graphs and Structural Equation Models
2. The Back Door Criterion
3. The Front Door Criterion
4. Bad Controls
5. Challenges for DAGs: Shape Restrictions
6. Challenges for DAGs: Non-recursive models
7. Challenges for DAGs: Big Models

DAG FOR COMPLETELY RANDOMIZED EXPERIMENT

- Randomized Experiment:



- We can also do this in a slightly different way where we make explicit the unobserved nodes:

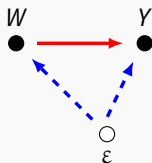


STRUCTURAL EQUATION MODEL

- The **structural equation model** form for this is:

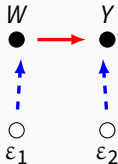
$$W = f_1(\varepsilon_1), \quad Y = f_2(W, \varepsilon_2)$$

- The key is that the first unobserved component ε_1 does not enter into the second equation. A model like the following would not work:



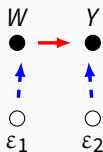
- The unobserved components are **independent** because there is no link between them.

TERMINOLOGY (1)



- ε_1 is a **parent** and **ancestor** of W
- W is a **child** and **descendant** of ε_1 , Y is a **child** and **descendant** of W and ε_2 .
- Y is a **descendant** of ε_1 (but not a **child**)
- ε_1 is an **ancestor** of Y (but not a **parent**)
- $\varepsilon_1, \varepsilon_2, W$ and Y are **nodes**.
- A **path** between two nodes is a set of connected **edges**, eg $\varepsilon_1 \rightarrow W \rightarrow Y \rightarrow \varepsilon_2$
- A **directed path** has all **arrows** going in the same **direction**, eg $\varepsilon_1 \rightarrow W \rightarrow Y$

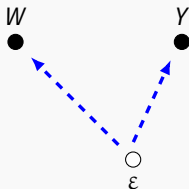
TERMINOLOGY (2)



- A **collider** is a **node** on a path with **arrows** going into the **node**, but not out of the **node**. Y is a **collider** on the **path** $\rightarrow W \rightarrow Y \rightarrow \varepsilon_2$
- A **backdoor path** from **node** A to **node** B is a **path** that starts with an **incoming node** into A and ends with an **incoming node** into B .
- A **path** between two **nodes** is **blocked** or **d-separated** by conditioning on a subset $\mathbb{Z}_1$ of the set of all nodes $\mathbb{Z}$ if and only if one of two conditions is satisfied.
 - Either (i) the path contains a **noncollider** that has been conditioned on,
 - or (ii) it contains a **collider** such that (a) that **collider** has not been conditioned on and (b) that **collider** has no **descendants** that have been conditioned on.

EXAMPLE II

- In this example there is no causal effect of W on Y , just an unobserved confounder leading to correlation.

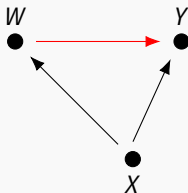


Structural Equation Model for this graph:

$$W = f_1(\varepsilon), \quad Y = f_2(\varepsilon)$$

THE BACK DOOR CRITERION

- Consider



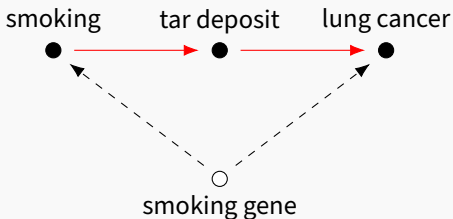
- (leaving out the unobserved nodes that only affect a single observed node).
- Structural Equation Model for this graph:

$$X = f_1(\varepsilon_1) \quad W = f_2(X, \varepsilon_2) \quad Y = f_3(X, W, \varepsilon_3)$$

BACK-DOOR CRITERION

- The **back-door criterion** for identifying the causal effect of a node W on a node Y is based on blocking **all** backdoor paths from W to Y through conditioning on a subset of nodes.
- Let us call this subset of conditioning variables $\mathbb{Z}$.
- When is it sufficient to condition on this subset?
- We need to check whether **all** back-door paths are blocked as a result of conditioning on $\mathbb{Z}$.
- Recall the definition of blocking or d-separating a backdoor path.
- It requires either conditioning on a non-collider, or the combination of not conditioning on a collider and not conditioning on all the descendants of that collider.
- In the last DAG, conditioning on X works: X is not a collider and blocking on it closes off the one back-door path $W \leftarrow X \rightarrow Y$.

THE FRONT DOOR CRITERION

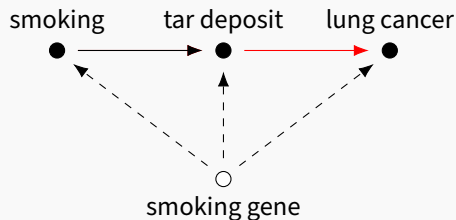


- Original Pearl DAG for front-door criterion.
- Note: we can not estimate the effect of smoking on lung-cancer directly because there is a back-door path through the smoking gene.
- We can't condition on the smoking gene because we do not observe it and we cannot directly estimate the effect of smoking on lung cancer because of the unobserved confounder smoking gene.

FRONT-DOOR CRITERION

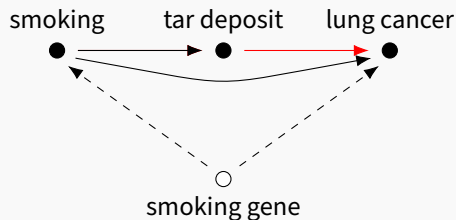
- A set of variables Z is said to satisfy the front-door criterion relative to an ordered pair of variables (X, Y) , if:
 - Z intercepts all directed paths from X to Y
 - There is no unblocked backdoor path from X to Z
 - All backdoor paths from Z to Y are blocked by X
- tar deposit satisfies these criteria

FRONT-DOOR CRITERION: POSSIBLE VIOLATIONS



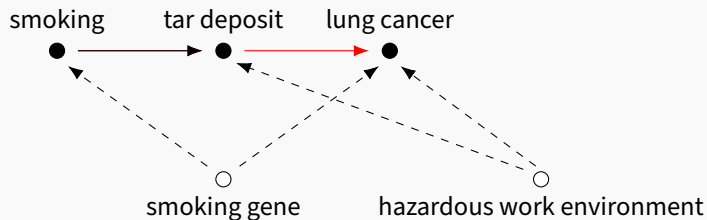
- Freedman Concern 1: smoking gene $\rightarrow$ tar deposits

FRONT-DOOR CRITERION: POSSIBLE VIOLATIONS



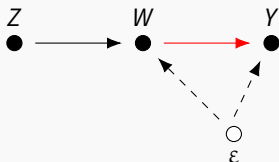
- Freedman Concern 2: $\text{smoking} \rightarrow \text{lung cancer}$

FRONT-DOOR CRITERION: POSSIBLE VIOLATIONS



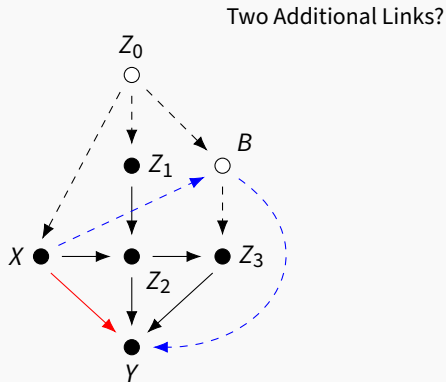
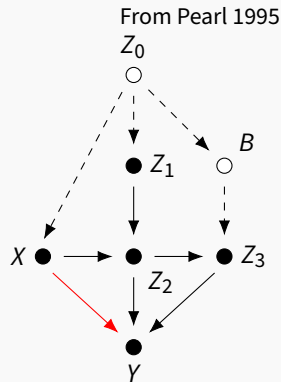
- Imbens Concern: hazardous work environment

INSTRUMENTAL VARIABLES



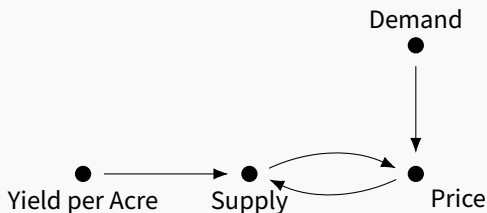
- Effect of W on Y is not identified (we knew that from a previous lecture).
- No obvious way to incorporate monotonicity in the graph
- No clear way of expressing **identification** of local average treatment effect.

PEARL 1995 DAG



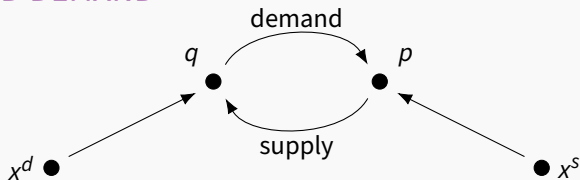
- X is fumigation, Y is yield, B is bird population, Z_t eelworm population: last season ($t = 0$), at beginning of season before fumigation ($t = 1$), after fumigation ($t = 2$) and end of season ($t = 3$), B bird population.
- How plausible is the DAG here?

HOW DO WE MODEL SUPPLY AND DEMAND IN A GRAPH? IS IT CYCLICAL?

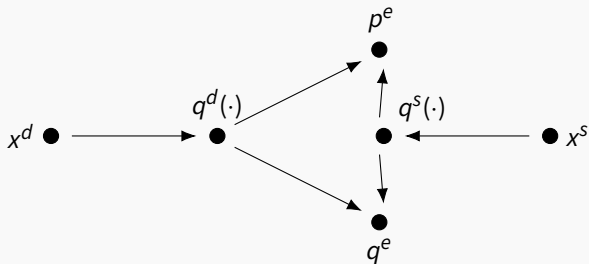


Based on Figure 7.10 in TBOW, p. 251.

SUPPLY AND DEMAND

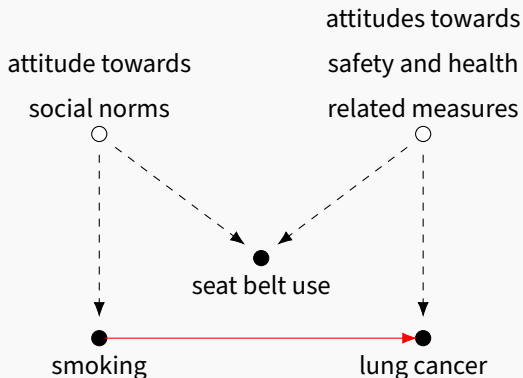


Demand and Supply I: Cyclical Version



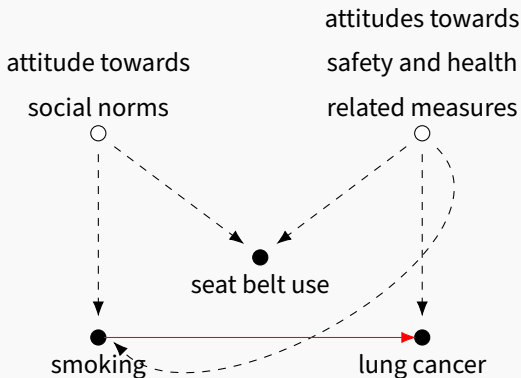
Demand and Supply II: Supply and Demand functions are nodes, Imbens, 2021, Journal of Economic Literature

BAD CONTROLS AND M-BIAS



M-Bias Assumption Satisfied: Conditioning on collider “seat belt use” introduces bias.

BAD CONTROLS AND M-BIAS



Violation of M-Bias Assumption

FURTHER DISCUSSION

- For further discussion between potential outcome approaches and directed acyclical graphs see
 - <https://causality.cs.ucla.edu/blog/index.php/2014/10/27/are-economists-smarter-than-epidemiologists-comments-on-imbenss-recent-paper/>
 - <https://causality.cs.ucla.edu/blog/index.php/2020/01/29/on-imbens-comparison-of-two-approaches-to-empirical-economics/>

REFERENCES

- FREEDMAN, DAVID A. "STATISTICAL MODELS FOR CAUSATION: WHAT INFERENTIAL LEVERAGE DO THEY PROVIDE?." *Evaluation review* 30, NO. 6 (2006): 691-713.
- IMBENS, GUIDO W. "POTENTIAL OUTCOME AND DIRECTED ACYCLIC GRAPH APPROACHES TO CAUSALITY: RELEVANCE FOR EMPIRICAL PRACTICE IN ECONOMICS." *Journal of Economic Literature* 58, NO. 4 (2020): 1129-1179.
- PEARL, JUDEA. "CAUSAL DIAGRAMS FOR EMPIRICAL RESEARCH." *Biometrika* 82, NO. 4 (1995): 669-688.
- PEARL, JUDEA. *Causality*. CAMBRIDGE UNIVERSITY PRESS, 2009
- PEARL, JUDEA, AND DANA MACKENZIE. *The book of why: the new science of cause and effect*. BASIC BOOKS, 2018.
- [HTTPS://CAUSALITY.CS.UCLA.EDU/BLOG/INDEX.PHP/2014/10/27/ARE-ECONOMISTS-SMARTER-THAN-EPIDEMIOLOGISTS-COMMENTS-ON-IMBENS-RECENT-PAPER/](https://causality.cs.ucla.edu/blog/index.php/2014/10/27/are-economists-smarter-than-epidemiologists-comments-on-imbenss-recent-paper/)
- [HTTPS://CAUSALITY.CS.UCLA.EDU/BLOG/INDEX.PHP/2020/01/29/ON-IMBENS-COMPARISON-OF-TWO-APPROACHES-TO-EMPIRICAL-ECONOMICS/](https://causality.cs.ucla.edu/blog/index.php/2020/01/29/on-imbens-comparison-of-two-approaches-to-empirical-economics/)